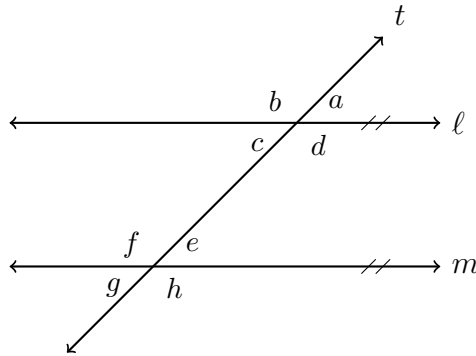


Skills Summary

Angle Measures on Parallel Lines

Use this reference while completing your worksheet or when studying.



With $\ell \parallel m$, every angle pair is one of two kinds.

- **Congruent (equal):** corresponding (a and e), alternate interior (c and e), alternate exterior (a and g), and vertical angles (a and c).
- **Supplementary (sum a straight angle):** any linear pair (a and b), and same-side interior angles (d and e).

Deciding which kind you have is the whole problem. Everything after it is arithmetic.

1. Find an angle from one you already know

Name the pair first, then act: congruent means copy the measure across; supplementary means subtract from a straight angle. When no single pair links the two angles, go through a third angle.

Example: $m\angle a = 130^\circ$. Find $m\angle e$ and $m\angle d$.

Step 1. $\angle e$ sits at line m where $\angle a$ sits at line ℓ , so they correspond and the parallel lines make them congruent.

Step 2. So $m\angle e$ copies across. For $\angle d$, note that $\angle a$ and $\angle d$ are a linear pair, so $180 - 130 = 50$.

$$m\angle e = 130^\circ \text{ and } m\angle d = 50^\circ.$$

Try it: $m\angle c = 68^\circ$. Find $m\angle g$ and $m\angle f$.

check: $m\angle g = 68^\circ$, $m\angle f = 112^\circ$

2. Solve for x when the angles are equal

Corresponding, alternate interior, alternate exterior and vertical pairs are congruent, so set the two expressions equal and solve. Then substitute back: the question almost always wants the angle, not x .

Example: $m\angle b = (2x + 25)^\circ$ and $m\angle f = (4x - 5)^\circ$. Find x and $m\angle b$.

Step 1. $\angle b$ and $\angle f$ are corresponding angles on parallel lines, so they are congruent — set the expressions equal rather than adding them.

Step 2. $2x + 25 = 4x - 5$ gives $25 + 5 = 4x - 2x$, so $x = 15$; then $2(15) + 25 = 55$.
 $x = 15$ and $m\angle b = 55^\circ$.

Try it: $m\angle d = (6x - 7)^\circ$ and $m\angle f = (3x + 20)^\circ$ are alternate interior angles. Find x and $m\angle d$.

check: $m\angle f = 65^\circ$, $m\angle d = 65^\circ$

3. Solve for x when the angles add to 180

A linear pair and a same-side interior pair are supplementary, so add the two expressions and set the sum to a straight angle. Same finish as before: substitute x back to get the measure.

Example: $m\angle d = (5x + 10)^\circ$ and $m\angle e = (3x + 2)^\circ$. Find x and $m\angle d$.

Step 1. $\angle d$ and $\angle e$ are same-side interior angles, the one parallel-line pair that is supplementary rather than congruent — so add, do not equate.

Step 2. $(5x + 10) + (3x + 2) = 180$, and solving gives $x = 21$; then $5(21) + 10 = 115$.
 $x = 21$ and $m\angle d = 115^\circ$.

Try it: $m\angle c = (7x + 4)^\circ$ and $m\angle f = (5x + 8)^\circ$ are same-side interior angles. Find x and $m\angle c$.

check: $m\angle f = 106^\circ$, $m\angle c = 106^\circ$

(!) Watch out: two habits cost the most marks. The first is treating every parallel-line pair as congruent — same-side interior pairs add to a straight angle instead. The second is answering with x when the question asked for an angle. A quick test: add your two angle measures and see whether the result matches the pair you claimed.